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May 22, 2026

Therapy Science Lab

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Subject: Application for PhD Position , Movement Biomarkers in Stroke Recovery (CEREBRIS / AxoMotris)

Dear Members of the Therapy Science Lab,

I built a motion capture system from scratch. I did it because the research question I was trying to answer needed it. A single camera system cannot and missed important information and I wanted to reduce injuries. I reconstruct 3D human movement from dual side video by picking the 2D keypoints and form 3D reconstruction. I had to check if it was accurate by comparing it to marker-based data. Then I used it to find problems that an expert would notice. What I found out was that the hardest part of markerless motion capture is not the algorithm. Its making a system that works well in a lab and also gives results in real life. The AxoMotris project is working to close this gap. That's why I am applying for the PhD position on Movement Biomarkers in Stroke Recovery, at the Lake Lucerne Institute.

My MPhil in Computer Science at the University of Haripur, Pakistan, was built entirely around multi-view markerless kinematics. My thesis developed a dual-camera pipeline for 3D human pose reconstruction: synchronized video capture, 2D keypoint detection using MediaPipe's deep learning backbone, and 3D reconstruction via Direct Linear Transformation across calibrated camera views. The system was validated against a Vicon marker-based motion capture reference , 15 subjects, over 4,000 annotated frames , and achieved a Mean Per Joint Position Error of 8.77 ± 1.72 mm. I also built a quality assurance framework for the kinematic dataset: inter-rater reliability assessment using Cohen's κ (0.84 for phase labelling, 0.79 for fault identification), one-way ANOVA with Tukey post-hoc testing across joint categories, and 95% confidence intervals computed using t -distribution across all validation metrics. A biomechanical fault detection module analysed joint angles against clinical thresholds and achieved 85.3% agreement with certified domain experts. A manuscript based on this work is currently under review at *Scientific Reports* (Springer Nature Journal the manuscript has successfully passed the first round of peer review and the subsequent revision stage. Following further feedback from the second reviewer,

the final revised manuscript has been submitted. Official publication is anticipated for June 2026.), with revisions submitted and a decision expected in June 2026. My undergraduate project on an automated billing system was granted a national patent by IGNITE Pakistan's ICT R&D fund.

The primary role in this position , contributing to the development and validation of multi-view and monocular markerless motion capture pipelines, and building a quality assurance framework for kinematic datasets , is a precise description of what my thesis delivered. That isn't a coincidence I'm reaching for; it's a direct match. What changes in the CEREBRIS context is what matters most: the movement being analysed is upper-limb stroke recovery, and the downstream consumer of the kinematic output is not a sports coach but a clinician making rehabilitation decisions. That shift in stakes changes the standard to which the system must be held. It demands the same rigour I applied in my validation work , but now with clinical interpretability, regulatory awareness, and genuine patient outcomes on the line.

That is the research direction I want to pursue. I am specifically drawn to LLUI's model of ubiquitous rehabilitation , aligning interventions with a patient's actual clinical state and movement data rather than fixed protocols. The CEREBRIS consortium's goal of reaching Technology Readiness Level 4 by project completion means this PhD isn't purely theoretical; it's building something that will be evaluated in real clinical settings. Working within a consortium of 14 European partners also means genuine interdisciplinary collaboration with clinicians, engineers, and translational researchers , the kind of environment where a computational background like mine has to translate its outputs into something a physiotherapist can actually use.

My technical stack is Python-first, with hands-on experience in PyTorch, TensorFlow, OpenCV, NumPy, scikit-learn, and Matplotlib for machine learning, computer vision, and statistical data analysis. I have direct experience building multi-view synchronisation pipelines, designing kinematic feature extraction modules, and validating system outputs against reference standards with full statistical reporting. I am comfortable with Git-based collaborative development, Linux environments, and structured documentation for research pipelines , relevant to the containerisation and version control practices the role notes as desirable. I have working knowledge of MATLAB, and my experience with real-time pipeline optimisation on GPU hardware applies directly to the performance requirements of a deployable markerless capture system. I understand what it means to build something not just for a paper but for eventual clinical use , where reliability, interpretability, and failure modes matter as much as accuracy metrics.

My long-term goal is to develop biomechanical assessment systems that work reliably outside the lab , systems where the clinical utility is built in from the design stage, not retrofitted afterward. The AxoMotris position offers exactly the right combination of technical depth, clinical grounding, and European research infrastructure to pursue that goal seriously. I am

applying with full commitment to the September 2026 start and to the interdisciplinary demands of the CEREBRIS project.

I have attached my CV, academic transcripts and diplomas, and publication list. Contact details for two referees are provided below. I am happy to share a preprint of the manuscript or a repository link to code from the thesis pipeline on request.

References:

- **Dr. Sikandar Ali** , MPhil Thesis Supervisor
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- **Dr. Tasleem Arif** , Research Collaborator
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Sincerely,
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